
DO AI WEATHER MODELS MISS EXTREMES? EVIDENCE FROM TEN MONTHS OF VERIFICATION AGAINST EUROPEAN WEATHER STATIONS

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ABSTRACT

AI weather models are often reported to underestimate extremes, but most evidence concerns deterministic regression models verified against reanalysis. We evaluate twelve physical and AI forecast models against ECMWF IFS using ten months of European station observations. The evaluation covers 10 m wind, 2 m temperature, solar radiation, and precipitation within regimes defined from a fixed ERA5 1991–2020 climatology. We find no uniform AI-specific deficit in the tails. Several AI models remain more accurate than IFS under extreme conditions, while others deteriorate markedly; comparable variation occurs among physical models. Every model nevertheless exhibits a common conditional-error pattern, overpredicting low observations and underpredicting high observations. Attenuation of extreme values therefore does not imply a uniform loss of relative skill: tail performance depends on the model, variable, and evaluation setting rather than on whether the forecast is produced by AI or physical numerical modelling.

1 INTRODUCTION

Data-driven weather models have achieved strong performance on standard forecast metrics, often matching or surpassing operational numerical weather prediction (NWP) models (Bi et al., 2023; Lam et al., 2023; Ben Bouallègue et al., 2024). However, many of these models produce deterministic forecasts and are trained using mean-squared error. Such models tend to regress towards the conditional mean, smoothing small-scale variability and underestimating extremes, as reported for peak winds and hot, windy, and record-breaking events (Olivetti & Messori, 2024a; Bonavita, 2024; Charlton-Perez et al., 2024; Olivetti & Messori, 2024b; Zhang et al., 2026).

Most existing evidence concerns deterministic regression models evaluated against reanalysis. Probabilistically trained models, generative ensembles, and high-resolution regional models have received far less attention, particularly when verified against weather stations, where errors can be substantially larger (Jin et al., 2024; Gupta et al., 2025; Schlueter & Wagner, 2025). Generative models are particularly relevant because they represent a distribution of possible outcomes rather than a single conditional estimate, and recent works have shown promising performance for weather extremes and high-resolution regional forecasting (Price et al., 2025; Pathak et al., 2026; Molinaro et al., 2024).

A second limitation is that underestimation is often attributed specifically to AI, even though also NWP models may underestimate the same observations. An AI model can therefore underestimate an extreme while remaining closer to it than physical models. The relevant question is not whether AI models underestimate extremes, but whether they are less accurate than NWP when extremes occur.

We test this using thirteen forecast models, including ECMWF IFS as the reference. Forecasts are evaluated over ten months against European synoptic, solar-radiation, and rain-gauge observations. The analysis covers 10 m wind, 2 m temperature, hourly solar radiation, and hourly precipitation at lead times up to 48 h.

Throughout the paper, “extreme” denotes an observation in the tail of a local ERA5 1991–2020 climatology conditioned on station, calendar day, and UTC hour. These regimes identify conditions

that are unusual for a particular location and time of year; they do not necessarily correspond to record-breaking or impact-defined events.

Our main contributions are:

- We extend station-based evaluation of weather extremes to deterministic regression models, a CRPS-trained global ensemble, and high-resolution regional generative models.
- We evaluate both the direction and magnitude of forecast errors, separating underestimation shared by AI and physical models from differences in their accuracy under the same observed conditions.
- We examine these properties across four surface variables, forecast leads, countries, and model classes, with additional categorical verification of heavy precipitation and source-defined case studies of record-breaking temperature events and major windstorms.

2 METHODS

We begin by describing first the forecast models and station observations. We then introduce the climatological regimes used to identify extremes, followed by the bias-correction and evaluation procedures.

2.1 MODELS

Let $u(t)$ denote the atmospheric state at time t . Weather forecasting amounts to approximating its evolution from an initial state $u(t_0)$:

$$\hat{u}_m(t_0 + \tau) = \mathcal{F}_m^\tau(u(t_0)), \quad (1)$$

where m denotes the forecast model and τ the lead time. NWP models approximate $\mathcal{F}_m^\tau$ by numerically integrating the governing equations on a discrete grid, whereas AI models learn this mapping from historical atmospheric states. The resulting forecast may be deterministic, consisting of a single state, or probabilistic, consisting of an ensemble

$$\left\{ \hat{u}_m^{(k)}(t_0 + \tau) \right\}_{k=1}^K.$$

We evaluate twelve forecast models against the ECMWF IFS high-resolution deterministic forecast, which serves as the reference (Table S1). The comparison spans NWP and AI models, global and regional domains, and deterministic and ensemble forecasts.

The NWP models are ECMWF ENS, NOAA GFS, DWD ICON Global, and DWD ICON-EU. The AI models comprise the global regression models ECMWF AIFS, Microsoft Aurora, and Jua EPT-2 Reasoning; the global ensembles ECMWF AIFS ENS and Jua EPT-2e; the regional generative ensembles Jua EPT-2 HRRR and EPT-2.1 Europa; and the solar-specialised regional model EPT-2.1 Helios (Lang et al., 2024a; Bodnar et al., 2025; Molinaro et al., 2025; Lang et al., 2024b; Siegenheim, 2025; Jua AI, 2026; Molinaro, 2026). AIFS ENS is trained with a continuous ranked probability score objective and learned noise injection and is evaluated as a 51-member ensemble. Related generative methodology is described by Molinaro et al. (2026; 2024).

Models are evaluated only for variables with compatible outputs. Precipitation is available from the physical models and from EPT-2 Reasoning, EPT-2e, EPT-2 HRRR, and EPT-2.1 Europa. AIFS, AIFS ENS and Aurora are restricted to wind and temperature because their radiation and precipitation outputs cannot be matched to the hourly verification data.

2.2 STATION OBSERVATIONS AND EVALUATION

Forecasts are verified against three complementary in situ datasets: operational surface observations for 10 m wind and 2 m temperature, solar-radiation observations for hourly global horizontal irradiance, and observations from 4,005 rain gauges for hourly precipitation. Contributing networks and data providers are listed in Appendix A.

The evaluation covers thirteen countries: DE, FR, GB, ES, IT, PL, NL, BE, CH, NO, AT, CZ, and DK. The precipitation network excludes Italy and Poland. Forecasts are bilinearly interpolated to observation locations (Molinaro et al., 2026). Precipitation is scored only at gauges within each model’s domain, and regional comparisons use a common gauge sample. None of the Jua models was trained or fine-tuned on these observations; historical station errors enter only through the causal bias correction described below.

Two evaluation tracks account for differences in availability and output cadence. Track A covers 1 September 2025 to 30 June 2026. Wind and temperature use 6-hourly leads from 6 to 48 h, matching the cadence of AIFS, AIFS ENS, and Aurora. Solar radiation and hourly precipitation use leads from 1 to 48 h. Models without compatible outputs are omitted for the corresponding variable, and ECMWF ENS contributes to precipitation only at its native 6-hourly leads.

Track B covers 1 March to 30 June 2026 and compares EPT-2.1 Europa, EPT-2 HRRR, and ICON-EU over a common hourly grid from 1 to 48 h. EPT-2.1 Helios is additionally included for solar radiation. Detailed availability by variable, country, and period is reported in the supplement.

2.3 OBSERVED-VALUE REGIMES

Regimes are defined independently of the evaluation period using a fixed ERA5 1991–2020 climatology (Hersbach et al., 2020). ERA5 fields are bilinearly interpolated from the 0.25° grid to each station. For each station, calendar day, and UTC hour, the reference sample comprises values from the same UTC hour within a cyclic ± 15 -day window over the 30-year period, yielding 930 values before variable-specific filtering. Percentile thresholds are computed separately from such reference sample.

Wind and temperature use all reference values, whereas solar radiation uses positive ERA5 GHI values only. For precipitation, values below 0.1 mm define a separate dry regime, and P50, P75, and P95 are computed from wet reference hours only. Complete definitions and meteorological labels are given in Tables S2 and S3. This local-climatology construction follows established practice for climate extremes (Zhang et al., 2011; Klein Tank et al., 2009; Perkins & Alexander, 2013).

Each verifying observation y is then assigned to a regime by comparison with the thresholds for its station, calendar day, and UTC hour.

2.4 BIAS CORRECTION

Persistent model-wide biases can dominate forecast errors and obscure how performance changes across observed-value regimes. To reduce their influence, we apply a causal rolling bias correction before computing skill. For wind, temperature, and daytime solar radiation, the correction is additive. Let $\mathcal{D}_{w,h}$ contain the matched errors at valid hour h during the four weeks preceding week w . For model m ,

$$\bar{b}_{m,w,h} = \frac{1}{|\mathcal{D}_{w,h}|} \sum_{(f_m,y) \in \mathcal{D}_{w,h}} (f_m - y), \quad \tilde{f}_m(t) = f_m(t) - \bar{b}_{m,w,h(t)}. \quad (2)$$

The estimate is pooled across European stations and does not depend on initialisation or lead. For ensemble models, the same shift is applied to every member.

An additive correction is unsuitable for zero-bounded, right-skewed precipitation. Let $\mathcal{P}_w$ contain forecasts and observations from the preceding four weeks, pooled across initialisations and leads. We apply

$$a_{m,w} = \text{clip}_{[0.33, 3]} \left(\frac{\sum_{\mathcal{P}_w} y}{\sum_{\mathcal{P}_w} f_m} \right), \quad \tilde{f}_m(t) = a_{m,w} f_m(t). \quad (3)$$

This multiplicative correction adjusts systematic amount biases while preserving zero forecasts. Both procedures use only prior information and do not alter observation-defined regime membership or force the conditional error within each regime to zero.

2.5 SKILL AND AGGREGATION

For model m and evaluation cell c , MAE skill relative to ECMWF IFS is

$$S_{m,c} = 1 - \frac{\text{MAE}_{m,c}}{\text{MAE}_{\text{IFS},c}}. \quad (4)$$

Positive values indicate lower MAE than IFS. Scores are computed from matched forecast–observation pairs using the common intersection of available initialisations and the four daily cycles at 00, 06, 12, and 18 UTC.

Ensemble models are evaluated through their ensemble means; the analysis therefore assesses point-forecast accuracy rather than calibration or spread. Results are reported at selected leads and over the complete lead grid of each comparison. Lower-cadence models contribute only at their native leads, without temporal interpolation.

Uncertainty in aggregated skill scores is quantified using a leave-one-month-out jackknife, with error bars showing one standard error. For heavy precipitation, we additionally treat observations at or above the local P95 threshold as binary events and report probability of detection, false-alarm ratio, critical success index, and frequency bias. Scores are pooled across gauges and forecast leads up to 48 h.

3 RESULTS

3.1 SKILL ACROSS THE OBSERVED DISTRIBUTION

Figure 1 shows that tail skill does not follow a uniform AI–physical ordering. EPT-2.1 Europa leads the wind comparison overall and remains competitive with ICON Global at gale force. EPT-2 HRRR and EPT-2.1 Europa perform most strongly for temperature, including in the heat regime. By contrast, AIFS and AIFS ENS remain below IFS in the wind and temperature tails despite the ensemble’s improved performance near the centre of the distribution. The largest heat deficit belongs to the physical GFS model.

Solar radiation follows a different pattern. EPT-2.1 Helios leads overall and under overcast and near-clear-sky conditions but falls below IFS under typical irradiance, where EPT-2 Reasoning performs most strongly. Its advantage is therefore concentrated in the tails rather than distributed uniformly across conditions.

For precipitation, the clearest gains occur in the middle of the wet-hour distribution (Figure 2). EPT-2 HRRR, EPT-2e, and EPT-2 Reasoning improve on IFS from moderate through high rain, whereas the physical global models remain below the reference. These gains narrow at the heaviest intensities. Europa follows a different pattern: its overall advantage is driven largely by lower errors during dry hours, while it falls below IFS once precipitation is observed.

Categorical verification provides a complementary evaluation of the heaviest precipitation. All models underpredict the frequency of these events. Complete detection results are reported in Figure S7 and Table S24.

3.2 SHARED CONDITIONAL-ERROR PATTERN

Despite their differences in relative skill, all models exhibit the same conditional-error structure (Figure 3): low observations are overpredicted and high observations underpredicted. The variation across regimes is larger than the spread between models and is present for all four variables. A common tendency towards the centre of the observed distribution therefore coexists with model-specific differences in relative accuracy.

Because the regimes are defined using the verifying observations, this pattern does not by itself show that the models systematically suppress extreme values. The same relationship can arise from limited predictability, unresolved local variability, or differences between point observations and gridded forecasts.

3.3 RECORD-BREAKING TEMPERATURE EVENTS AND MAJOR WINDSTORMS

To complement the climatological-regime analysis, we evaluate five externally documented record-breaking temperature events and seven major European windstorms. Event windows and affected-country footprints were specified from external reports before scoring. A forecast sample is retained

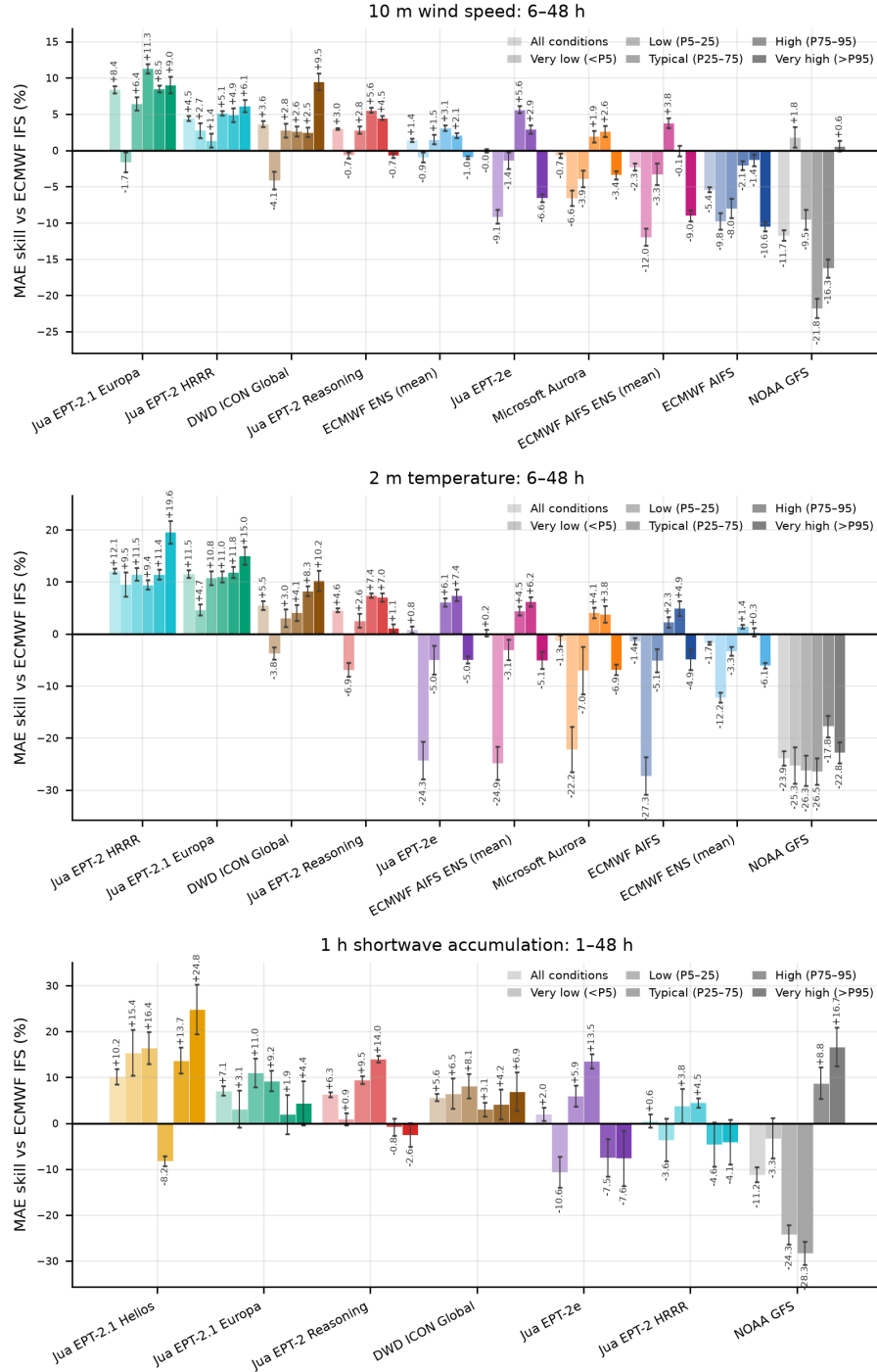


Figure 1: Track A MAE skill relative to ECMWF IFS by observed-value regime, with ± 1 jackknife SE. Wind and temperature use 6–48 h leads; solar radiation uses 1–48 h.

only when IFS and every Track A model included in the comparison share the same station, valid time, initialisation, and 6-hourly lead from 6 to 48 h. Each model therefore uses identical event windows and forecast–observation samples. Full event definitions, provenance, scoring details, and numerical results are reported in Appendix A.6.

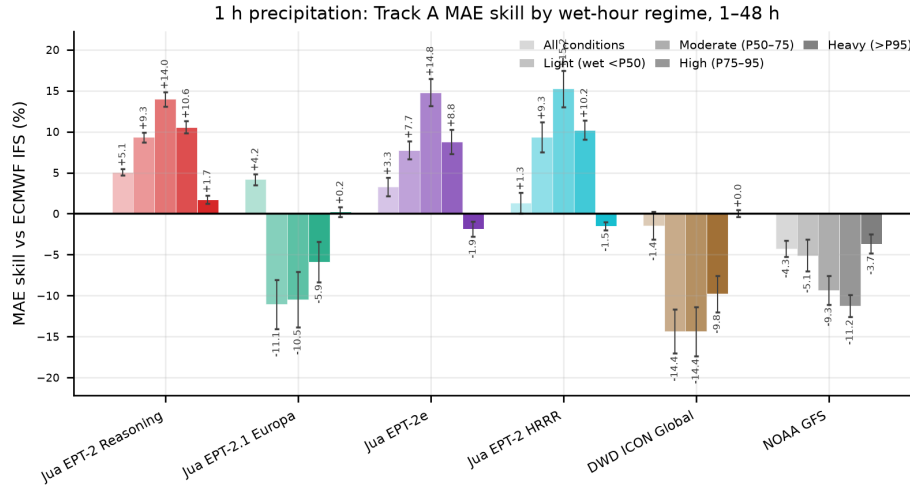


Figure 2: Track A MAE skill relative to ECMWF IFS for hourly precipitation by observed-value regime over 1–48 h, with ± 1 jackknife SE.

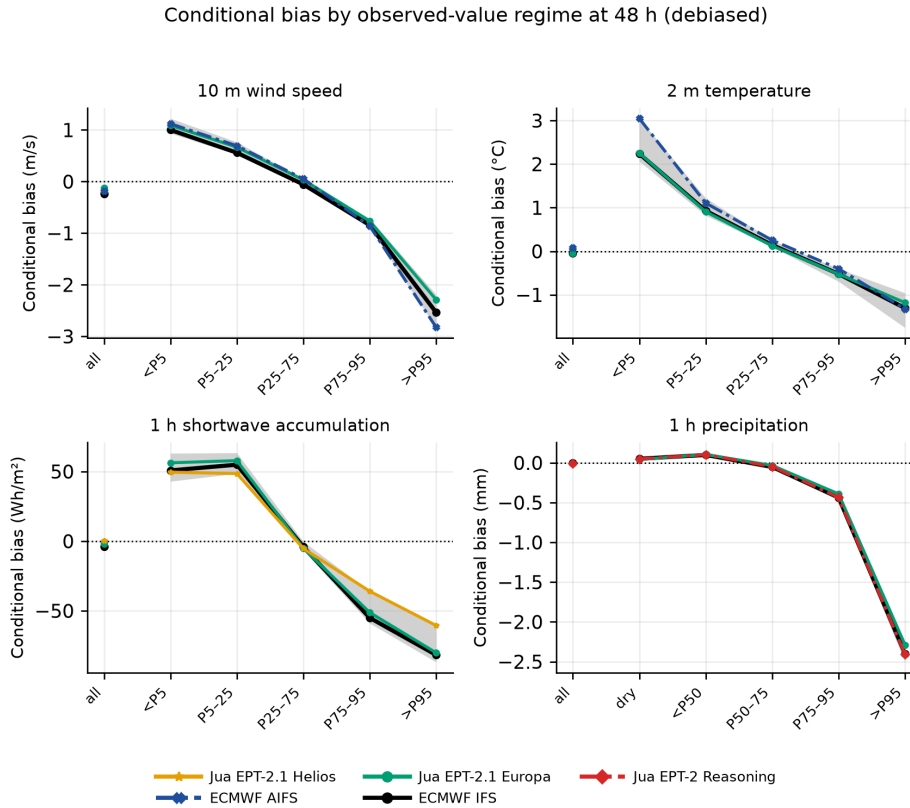


Figure 3: Mean forecast-minus-observation error at 48 h, conditioned on the observed-value regime after bias correction. The grey envelope spans all models; selected models are highlighted.

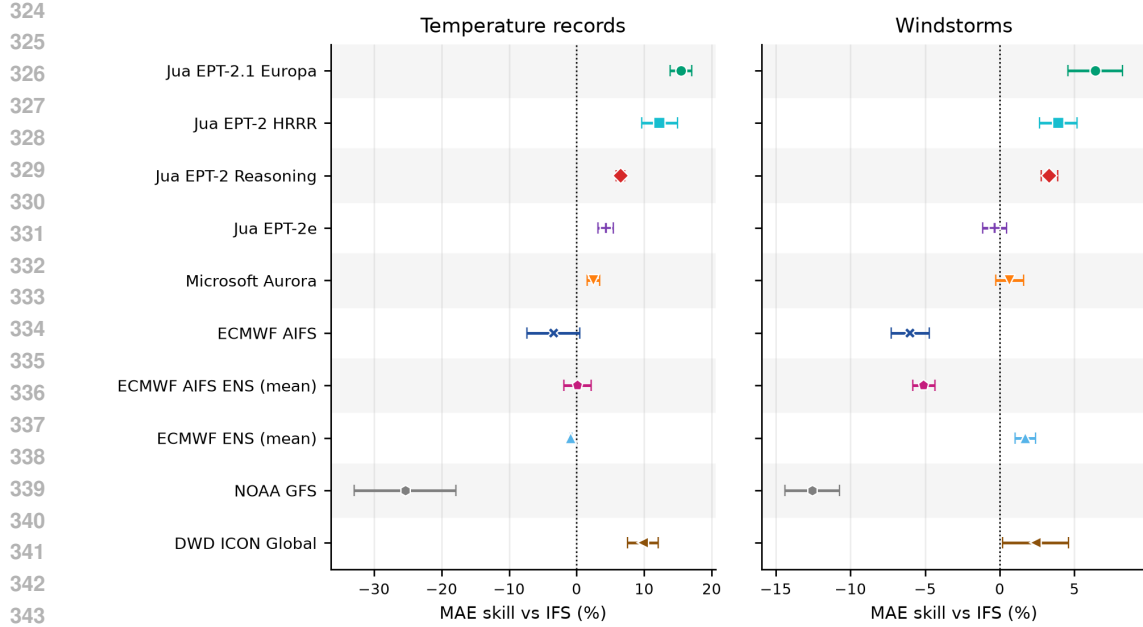


Figure 4: MAE skill relative to ECMWF IFS during five documented record-breaking temperature events (left; $n = 231,065$) and seven major European windstorms (right; $n = 123,847$), aggregated over common station/valid-time/lead samples from 6–48 h after bias correction. Error bars show one leave-one-event-out jackknife SE.

Europe shows positive skill in both event sets (Figure 4). During the temperature events, performance varies substantially across models, with GFS showing the largest deficit. During the windstorms, most models remain close to IFS, whereas AIFS, AIFS ENS, and GFS remain below the reference. These source-defined cases therefore support the model-specific pattern found in the climatological-regime analysis, rather than a uniform AI-specific loss of skill during extremes.

3.4 REGIONAL COMPARISON

Track B compares EPT-2.1 Europa and EPT-2 HRRR directly with ICON-EU over March–June 2026 (Figure 5). Europa retains the strongest overall wind estimate, while ICON-EU is competitive with the leading AI models in the wind and temperature tails. EPT-2.1 Helios has the clearest solar advantage.

Supplementary analyses show that these conclusions persist across lead times, short-range forecasts, and individual countries (Appendix A), which also reports complete precipitation, uncertainty, and categorical-verification results.

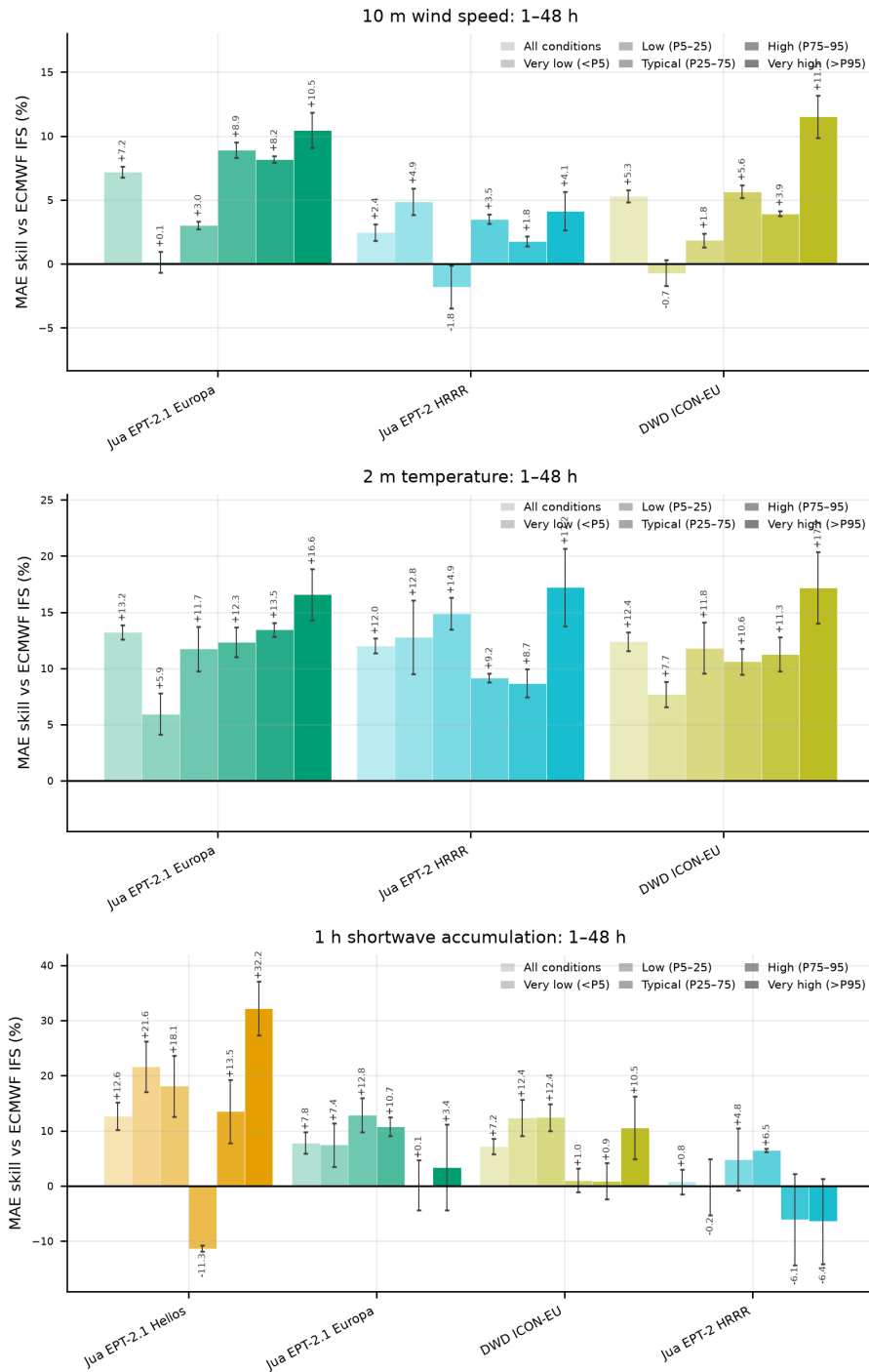


Figure 5: Track B MAE skill relative to ECMWF IFS by observed-value regime, aggregated over 1–48 h from March to June 2026, with ± 1 jackknife SE. Wind and temperature compare EPT-2.1 Europa, EPT-2 HRRR, and ICON-EU; the solar panel additionally includes EPT-2.1 Helios. Models are ordered by all-conditions skill within each panel.

4 DISCUSSION

The results distinguish between two questions: whether a model underestimates an extreme observation, and whether it is less accurate than IFS when that observation occurs. Across the evaluated models, low observations are overpredicted and high observations are underpredicted, although the magnitude of these errors varies. Several AI models are nevertheless closer to the observations than IFS in the heat, gale, solar, and precipitation regimes. Underestimation is therefore common to both AI and NWP models, whereas relative accuracy depends on the model, variable, and observed regime.

The same conditional-error pattern appears across deterministic, probabilistic, and generative AI models, as well as NWP models, with no consistent ordering between these classes. Instead, differences emerge at the level of individual systems: AIFS and AIFS ENS perform worse than IFS in the wind and temperature tails, whereas EPT-2 HRRR and EPT-2.1 Europa remain competitive at gale force and in heat. The NWP results likewise vary, with GFS showing the largest heat deficit while ICON-EU is competitive with the leading regional AI models in the matched Track B comparison. The source-defined temperature events and windstorms preserve this model-specific ordering on common event samples: the leading regional systems remain ahead of IFS, whereas the two AIFS variants and GFS lose skill in at least one event set.

Relative accuracy also depends on the observed regime and variable. For example, EPT-2.1 Helios is more accurate than IFS in both solar tails but less accurate under typical irradiance. Similarly, EPT-2 HRRR, EPT-2e, and EPT-2 Reasoning improve on IFS for moderate to high precipitation, although their advantage narrows for the heaviest events.

Within these scope conditions, the results do not support a general weakness of AI weather forecasting at extremes.

4.1 LIMITATIONS

The comparisons identify differences in performance, but not their causes. The systems differ in architecture, training objective, resolution, training data, initialisation, cadence, and domain (Bonavita, 2024; Subich et al., 2025); consequently, their performance cannot be attributed independently to AI versus NWP, regression versus generative training, or global versus regional resolution. Moreover, evaluating ensembles through their means assesses point-forecast accuracy, rather than calibration, spread, or event probabilities.

ETHICS STATEMENT

Several of the forecast systems evaluated in this study are developed by the authors' organisation. The verification protocol is applied identically across AI and physical systems, every comparison is reported against the independent ECMWF IFS reference, and the archived requests, aggregates and reproduction code described below are provided to support independent scrutiny of the comparisons. The study uses forecast output and meteorological station observations only; it involves no human subjects or personal data, and no station identities or per-station series are released.

REPRODUCIBILITY STATEMENT

Metrics were computed through a content-addressed station-verification API; requests and responses are archived. The reproduction package includes the aggregates used for the reported numbers, headline skill tables, and the extraction, aggregation and figure code. All score-based figures and tables regenerate offline from those aggregates. The package is provided as anonymised supplementary material.

AI USE STATEMENT

Generative AI tools were used to assist with software changes in the extraction, aggregation and figure code, to inspect generated results, and to draft and edit manuscript text. They were not used to generate or modify observations or forecast data. The authors checked AI-assisted code through

the pipeline’s executable self-tests and regenerated artifacts, verified the quoted numbers against the aggregate data, reviewed all manuscript text, and take full responsibility for the content.

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A SUPPLEMENTARY MATERIAL

This supplement provides the model and regime definitions omitted from the main text, lead-time and short-range robustness checks for Track A, the matched regional comparison in Track B, complete numerical tables, and country-level regime profiles.

A.1 EVALUATION SETUP

Surface wind and temperature observations were obtained through Synoptic Data from selected operational observing networks, including SYNOP, WIS2, ASOS/AWOS, and METAR. Solar-radiation observations were compiled from ARPALomb, BSRN, DMI, DWD, FMI, Frost, GeoSphere Austria, KNMI, Météo-France, MeteoSwiss, RMI, and SMHI.

Table S1 identifies the forecasting systems and their broad model classes. Tables S2 and S3 give the observed-value intervals and meteorological labels used throughout the evaluation.

Table S1: Forecast models included in the evaluation. ECMWF IFS is the reference model.

Model	Model class	Forecast type	Domain
ECMWF IFS (reference)	Physical	Deterministic	Global
ECMWF ENS	Physical	Ensemble	Global
NOAA GFS	Physical	Deterministic	Global
DWD ICON Global	Physical	Deterministic	Global
DWD ICON-EU	Physical	Deterministic	Regional
ECMWF AIFS	Regression AI	Deterministic	Global
Microsoft Aurora	Regression AI	Deterministic	Global
Jua EPT-2 Reasoning	Regression AI	Deterministic	Global
Jua EPT-2e	Regression AI	Ensemble	Global
Jua EPT-2.1 Helios	Regression AI (solar)	Deterministic	Regional
ECMWF AIFS ENS	Probabilistic AI	Ensemble	Global
Jua EPT-2 HRRR	Generative AI	Ensemble	Regional
Jua EPT-2.1 Europa	Generative AI	Ensemble	Regional

Table S2: Observed-value regimes for wind, temperature, and solar radiation. For solar radiation, the percentiles refer to positive daytime GHI.

Regime	Definition	Wind	Temperature	Solar
All	–	all conditions	all conditions	all daytime
Very low	$y < P5$	calm	severe cold	heavily overcast
Low	$P5 \leq y < P25$	light air	cold	overcast
Typical	$P25 \leq y < P75$	moderate	moderate	mixed sky
High	$P75 \leq y < P95$	fresh to strong	warm	mostly sunny
Very high	$y \geq P95$	gale force	heat	near clear sky

Table S3: Observed-value regimes for hourly precipitation. Wet-hour percentiles are conditional on ERA5 accumulations of at least 0.1 mm.

Regime	Definition	Interpretation
All	–	all hours
Dry	$y < 0.1$ mm	dry hour
Light	$0.1 \text{ mm} \leq y < P50$	light rain
Moderate	$P50 \leq y < P75$	moderate rain
High	$P75 \leq y < P95$	heavy rain
Heavy	$y \geq P95$	most intense hours

A.2 TRACK A LEAD-TIME, SHORT-RANGE, AND PRECIPITATION ROBUSTNESS

Figures S1–S4 show skill as a function of forecast lead for all conditions and the relevant upper or lower tail. Figure S5 repeats the regime comparison over 1–12 h, and Table S4 reports heat-regime skill by country. Figures 2–S7 and Tables S5 and S12 provide the full-horizon, short-range, and categorical precipitation checks.

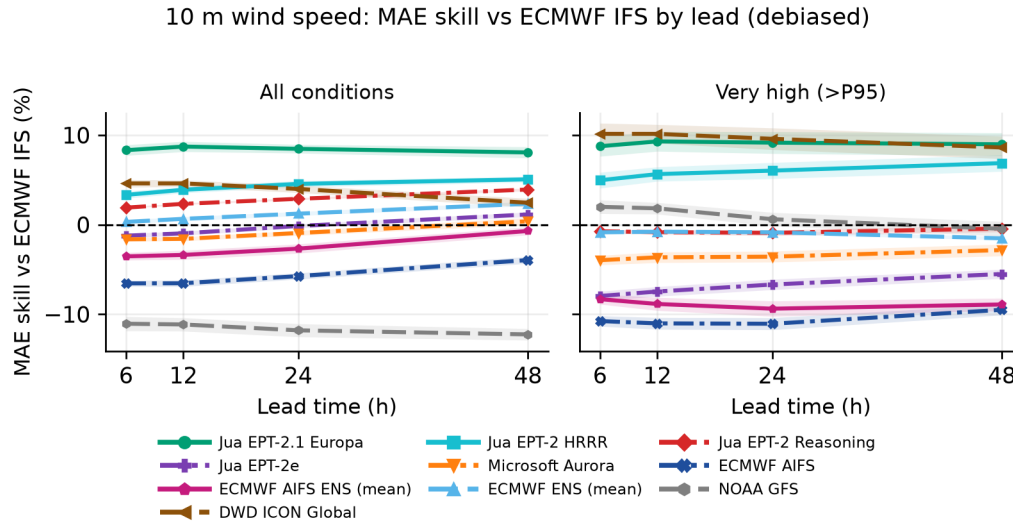


Figure S1: MAE skill relative to ECMWF IFS by lead time for 10 m wind. The left panel shows all conditions; the right panel shows the gale-force regime, $y \geq P95$. Shading denotes ± 1 jackknife SE.

Table S4: Track A MAE skill (%) by country in the heat regime ($y \geq P95$), aggregated over 6–48 h after bias correction. NOAA GFS is negative in every country with available observations.

Model	DE	FR	GB	ES	IT	PL	NL	BE	CH	NO	AT	CZ	DK
Jua EPT-2.1 Europa	+6.6	+16.1	-1.2	+6.4	+17.2	+6.2	-3.4	-3.0	+31.1	+5.5	+28.2	+16.6	-4.4
Jua EPT-2 HRRR	+10.5	+21.5	+4.0	+10.5	+27.6	+5.6	+1.8	-0.6	+29.4	+14.2	+16.6	+17.7	+12.2
Jua EPT-2 Reasoning	+6.6	+2.6	+4.3	-2.6	+0.1	+3.9	+3.8	-1.0	-1.3	-5.2	-3.1	+7.5	+2.7
Jua EPT-2e	+6.1	-2.2	-0.3	-15.1	-10.4	-2.4	+0.6	-2.8	-2.5	-12.2	-16.3	-1.3	+1.7
Microsoft Aurora	+2.1	-4.5	-7.5	-18.7	-7.8	-0.9	+0.7	-11.8	-3.3	-18.0	-15.4	-7.0	+1.8
ECMWF AIFS	+0.8	-1.6	-10.2	-12.7	-14.6	+1.9	-0.4	+6.3	+1.9	-3.4	-25.0	-3.1	+10.3
ECMWF AIFS ENS (mean)	+5.3	-3.1	-6.7	-10.6	-12.6	-3.1	-3.9	+5.1	+4.6	-3.2	-38.7	+2.1	+4.4
ECMWF ENS (mean)	-2.1	-4.6	-5.1	-12.1	-4.0	-8.6	+3.4	-2.9	-10.7	-9.8	-10.6	-1.5	-9.2
NOAA GFS	-18.4	-23.6	-37.0	-25.1	-13.2	-24.3	-21.9	-12.2	-10.2	-43.1	-25.3	-31.4	-24.7
DWD ICON Global	+6.9	+8.8	-4.5	+3.6	+12.8	+4.8	-2.7	+10.0	+22.7	+18.1	+16.8	+15.5	-6.5
DWD ICON-EU [†]	+13.1	+18.4	+1.3	+7.4	+20.1	+9.1	-2.6	+14.1	+34.6	+6.6	+28.1	+17.8	-0.7

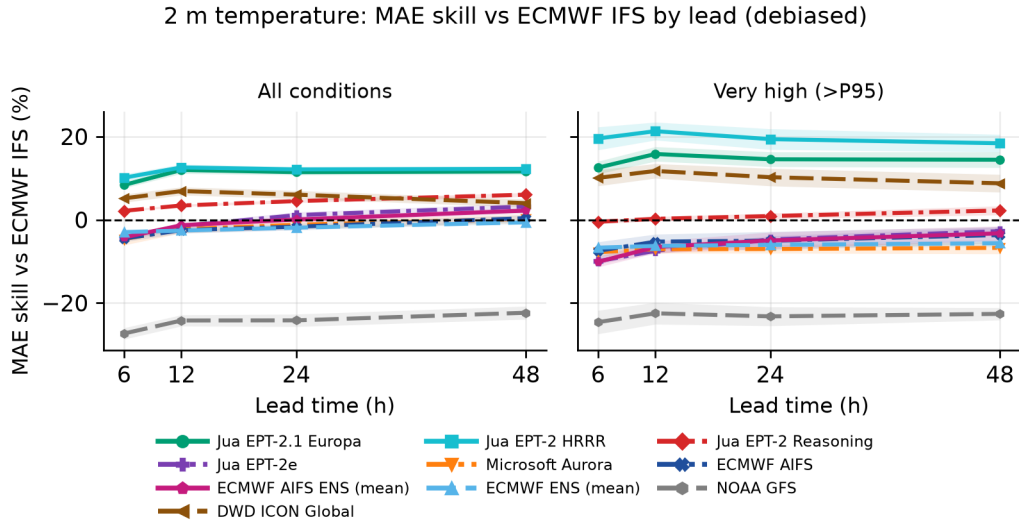


Figure S2: As Figure S1, for 2 m temperature. The right panel shows the heat regime, $y \geq P95$.

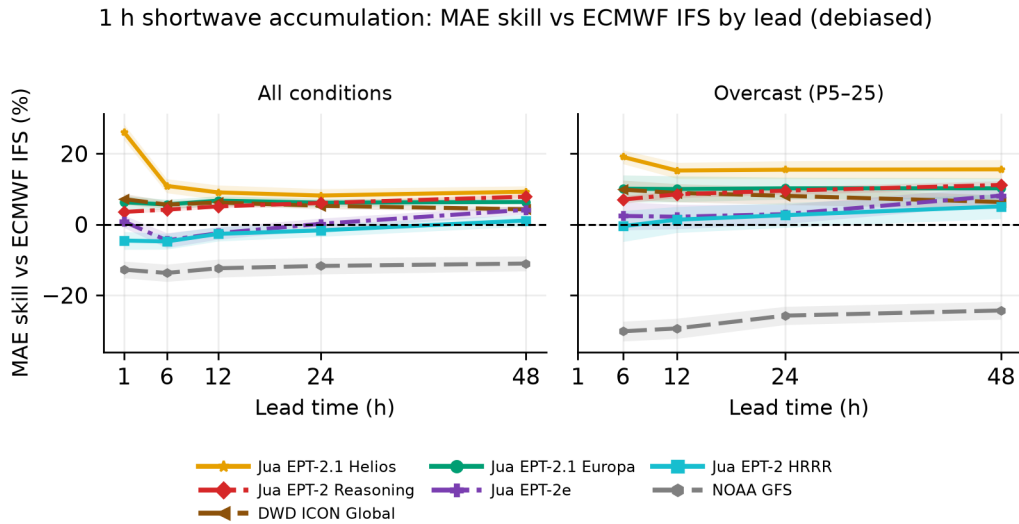


Figure S3: Track A MAE skill relative to ECMWF IFS by lead time for hourly solar radiation. The left panel shows all daytime conditions; the right panel shows the overcast regime, $P5 \leq y < P25$. AIFS, AIFS ENS, Aurora, and ECMWF ENS are omitted because they lack compatible hourly solar output. Shading denotes ± 1 jackknife SE.

Table S5: Track A MAE skill (%) relative to ECMWF IFS for hourly precipitation by observed-value regime, aggregated over 1–48 h after multiplicative bias correction. Parentheses give the leave-one-month-out jackknife SE.

Model	All	Dry	<P50	P50–75	P75–95	>P95
Jua EPT-2 Reasoning	+5.1 (± 0.4)	+2.2 (± 1.3)	+9.3 (± 0.6)	+14.0 (± 0.9)	+10.6 (± 0.8)	+1.7 (± 0.5)
Jua EPT-2 HRRR	+1.3 (± 1.3)	-6.1 (± 3.3)	+9.3 (± 1.8)	+15.2 (± 2.2)	+10.2 (± 1.2)	-1.5 (± 0.5)
Jua EPT-2e	+3.3 (± 1.1)	+1.5 (± 4.8)	+7.7 (± 1.1)	+14.8 (± 1.7)	+8.8 (± 1.5)	-1.9 (± 0.9)
Jua EPT-2.1 Europa	+4.2 (± 0.7)	+21.0 (± 2.7)	-11.1 (± 3.0)	-10.5 (± 3.4)	-5.9 (± 2.5)	+0.2 (± 0.6)
NOAA GFS	-4.3 (± 1.0)	-0.2 (± 1.8)	-5.1 (± 2.0)	-9.3 (± 1.8)	-11.2 (± 1.4)	-3.7 (± 1.2)
DWD ICON Global	-1.4 (± 1.7)	+8.3 (± 2.7)	-14.4 (± 2.7)	-14.4 (± 3.0)	-9.8 (± 2.2)	+0.0 (± 0.4)

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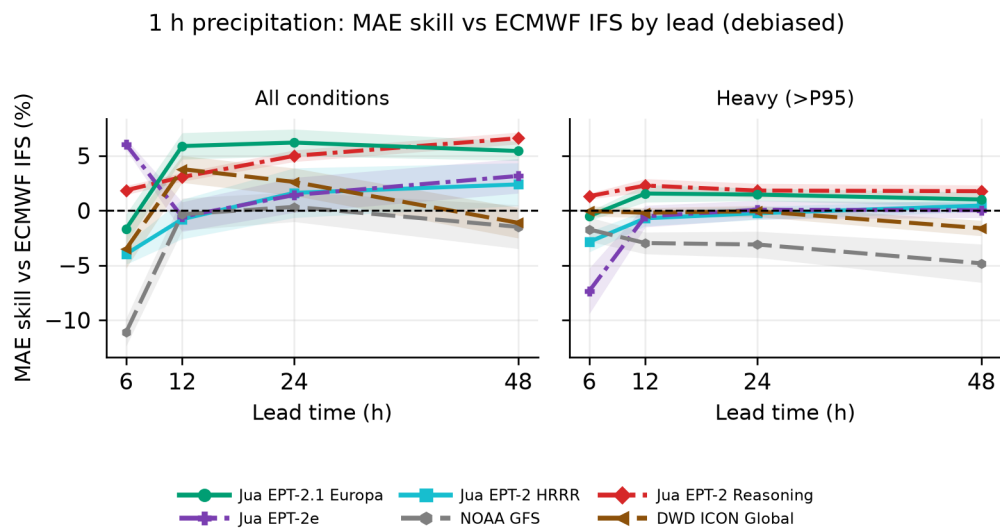


Figure S4: Track A MAE skill relative to ECMWF IFS by lead time for hourly precipitation. The left panel shows all conditions; the right panel shows the heavy regime, $y \geq P95$. Shading denotes ± 1 jackknife SE.

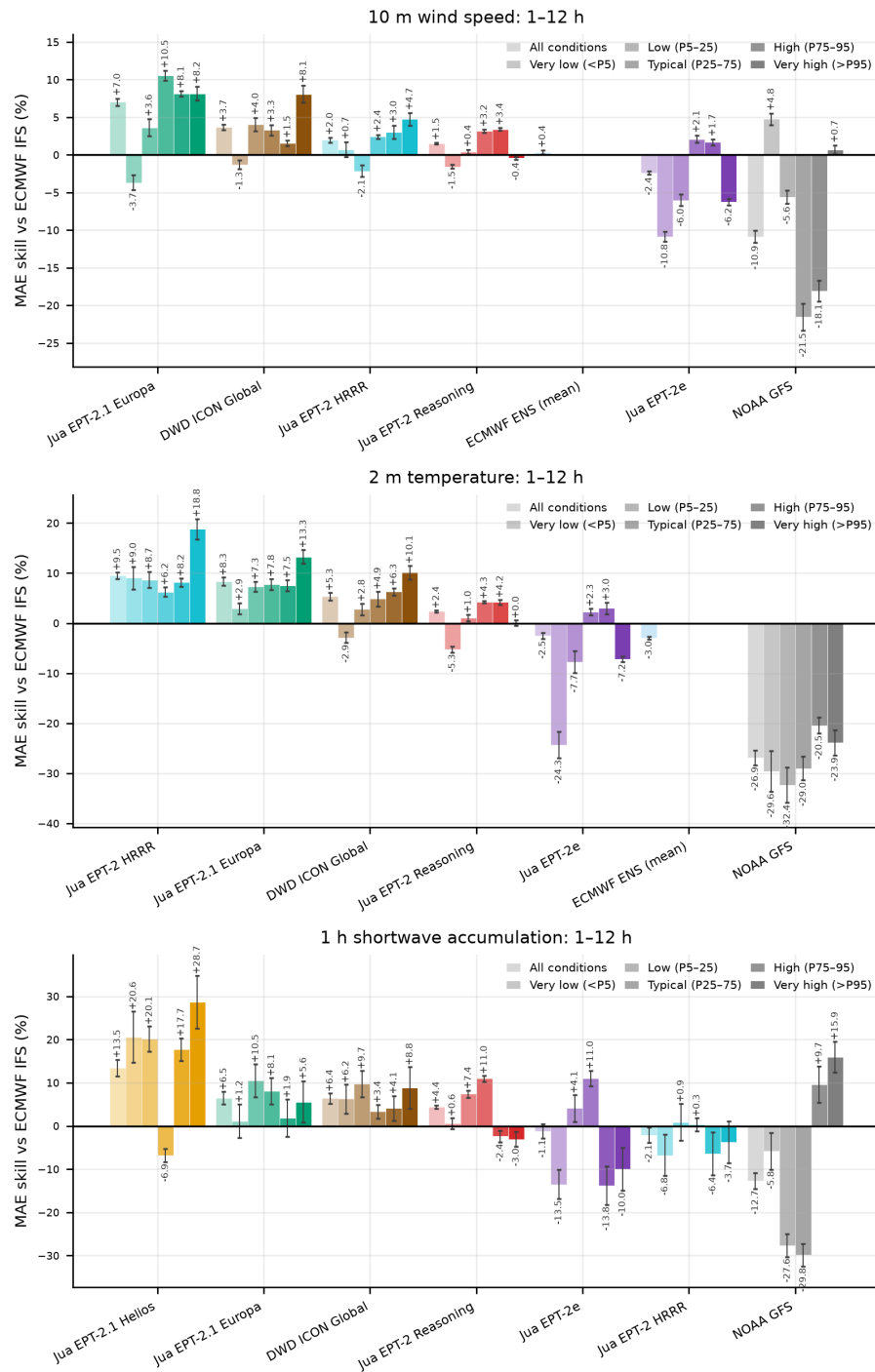


Figure S5: Track A MAE skill relative to ECMWF IFS by observed-value regime over 1–12 h, with ± 1 jack-knife SE. Wind and temperature are restricted to models with hourly output; solar radiation includes EPT-2.1 Helios.

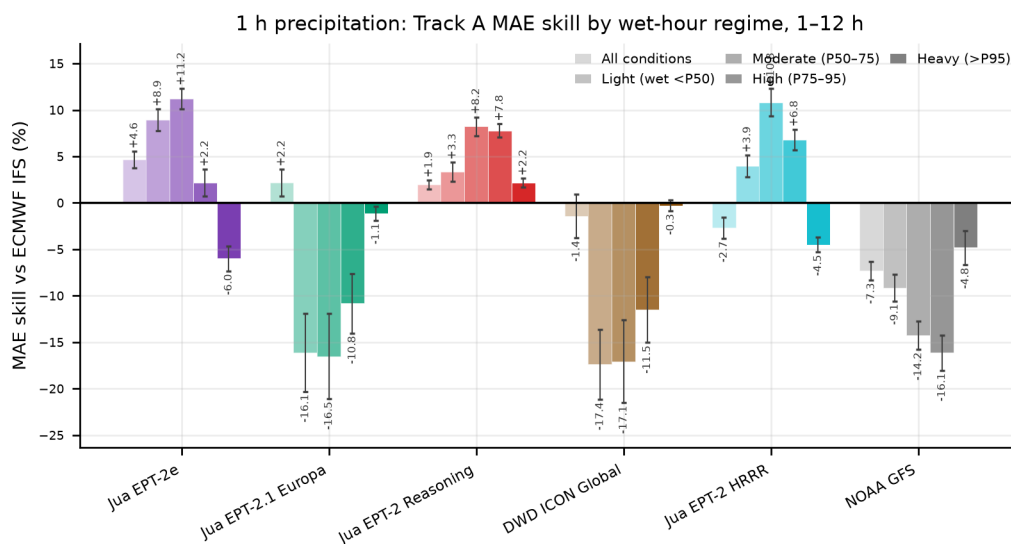


Figure S6: As Figure 2, restricted to the 1–12 h forecast range.

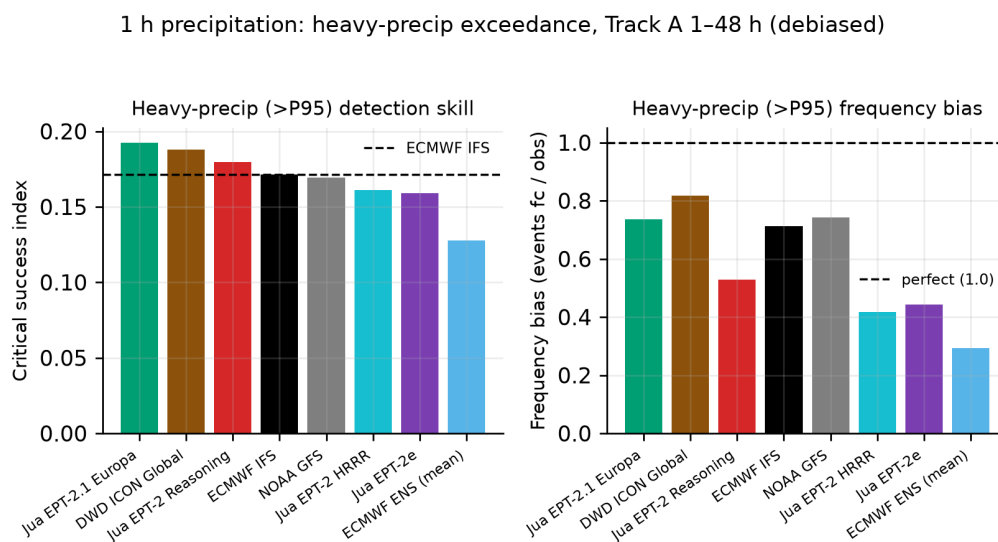


Figure S7: Detection of heavy precipitation ($y \geq P95$) on Track A over 1–48 h. Left: critical success index, with ECMWF IFS shown by the dashed line. Right: frequency bias, with unity denoting the correct event frequency. All models underpredict the frequency of heavy precipitation.

A.3 MATCHED REGIONAL COMPARISON

Track B robustness beyond the full-horizon comparison in the main text is shown in Figure S8 for 1–12 h and in Figure S9 as a function of lead. Figures S10 and S11 give the corresponding precipitation results.

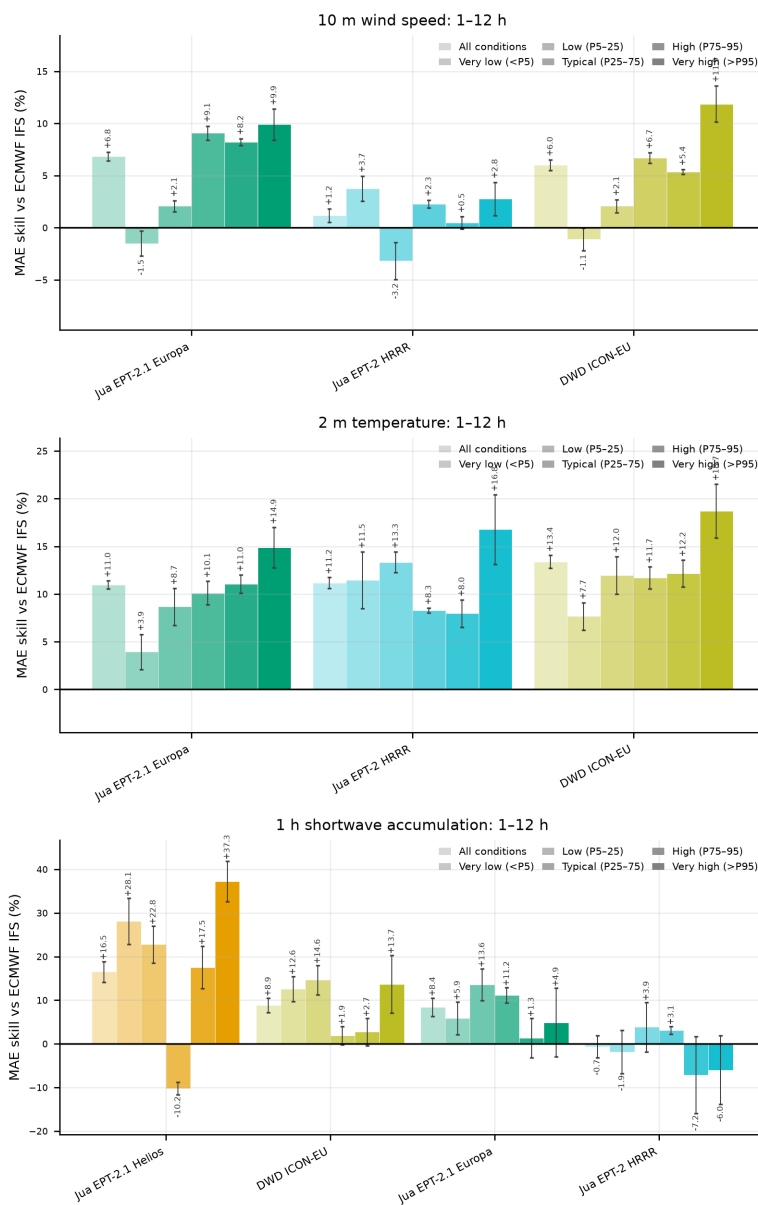


Figure S8: As Figure 5, restricted to the 1–12 h forecast range.

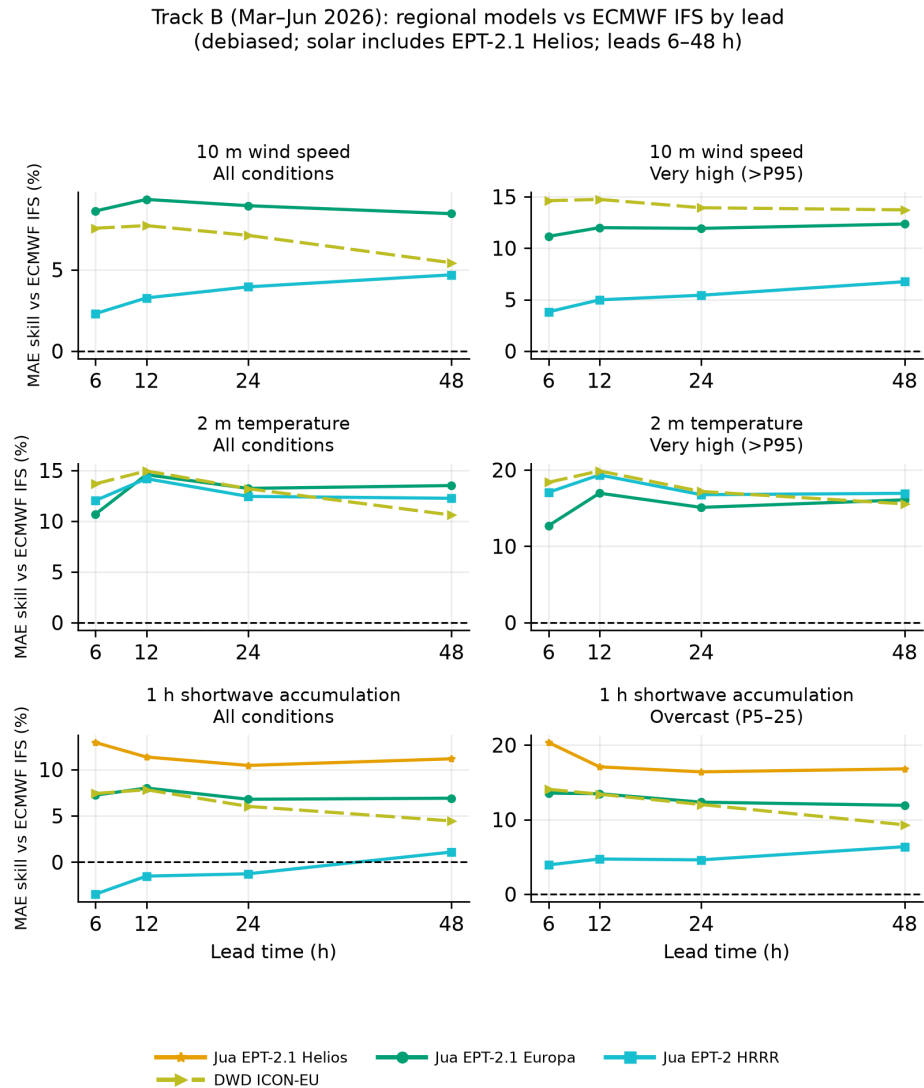


Figure S9: Track B MAE skill relative to ECMWF IFS by lead time. Rows show 10 m wind, 2 m temperature, and hourly solar radiation. Columns compare all conditions with gale-force wind, heat, and overcast conditions, respectively. Solar radiation additionally includes EPT-2.1 Helios. Plotted leads are 6, 12, 24, and 48 h; shading denotes ± 1 jackknife SE.

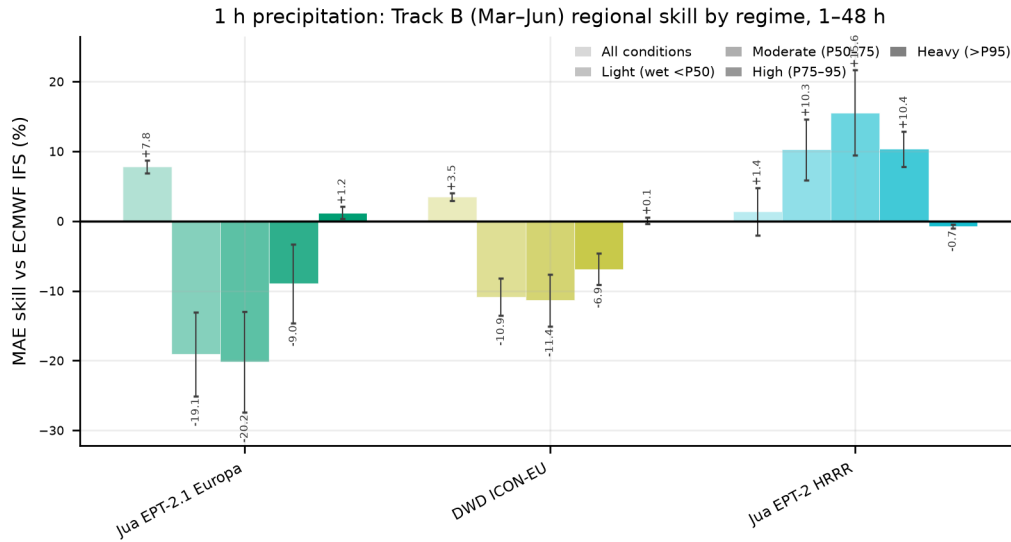


Figure S10: Track B MAE skill relative to ECMWF IFS for hourly precipitation by observed-value regime, aggregated over 1–48 h from March to June 2026, with ± 1 jackknife SE.

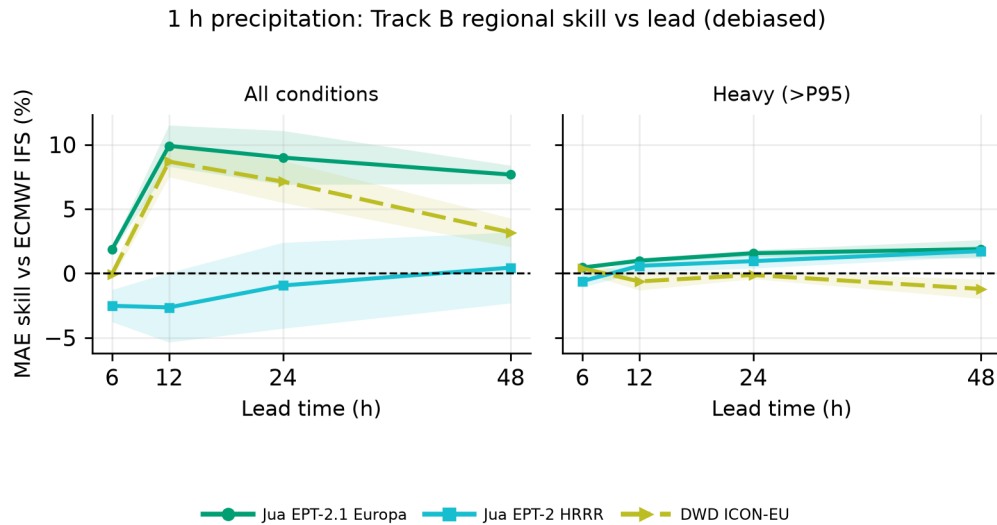


Figure S11: Track B precipitation skill relative to ECMWF IFS by lead time. The left panel shows all conditions; the right panel shows the heavy regime, $y \geq P95$. Shading denotes ± 1 jackknife SE.

A.4 COMPLETE NUMERICAL RESULTS

Tables S6–S15 report Track A regime, short-range, and lead-time values. Tables S16–S22 report the corresponding Track B values; Tables S23 and S24 complete the regional precipitation and heavy-event detection results.

Table S6: Track A MAE skill (%), 6–48 h, by regime – Figure 1, wind. Parentheses: jackknife SE.

Model	All	<P5	P5–25	P25–75	P75–95	>P95
Jua EPT-2.1 Europa	+8.4 (0.5)	-1.7 (1.4)	+6.4 (0.9)	+11.3 (0.7)	+8.5 (0.4)	+9.0 (1.2)
Jua EPT-2 HRRR	+4.5 (0.3)	+2.7 (1.0)	+1.4 (0.9)	+5.1 (0.3)	+4.9 (1.0)	+6.1 (0.9)
Jua EPT-2 Reasoning	+3.0 (0.1)	-0.7 (0.4)	+2.8 (0.5)	+5.6 (0.3)	+4.5 (0.3)	-0.7 (0.3)
Jua EPT-2e	-0.0 (0.2)	-9.1 (1.0)	-1.4 (1.1)	+5.6 (0.5)	+2.9 (0.6)	-6.6 (0.5)
Microsoft Aurora	-0.7 (0.3)	-6.6 (1.1)	-3.9 (1.1)	+1.9 (0.8)	+2.6 (0.8)	-3.4 (0.6)
ECMWF AIFS	-5.4 (0.4)	-9.8 (1.2)	-8.0 (1.3)	-2.1 (0.6)	-1.4 (0.8)	-10.6 (0.6)
ECMWF AIFS ENS (mean)	-2.3 (0.5)	-12.0 (1.2)	-3.3 (1.5)	+3.8 (0.7)	-0.1 (0.7)	-9.0 (0.7)
ECMWF ENS (mean)	+1.4 (0.2)	-0.9 (0.7)	+1.5 (0.6)	+3.1 (0.4)	+2.1 (0.4)	-1.0 (0.2)
NOAA GFS	-11.7 (0.7)	+1.8 (1.4)	-9.5 (1.4)	-21.8 (1.4)	-16.3 (1.3)	+0.6 (0.7)
DWD ICON Global	+3.6 (0.4)	-4.1 (1.2)	+2.8 (1.0)	+2.6 (0.7)	+2.5 (0.7)	+9.5 (1.1)

Table S7: Track A MAE skill (%), 6–48 h – Figure 1, temperature.

Model	All	<P5	P5–25	P25–75	P75–95	>P95
Jua EPT-2.1 Europa	+11.5 (0.7)	+4.7 (1.0)	+10.8 (1.3)	+11.0 (1.0)	+11.8 (1.1)	+15.0 (1.7)
Jua EPT-2 HRRR	+12.1 (0.4)	+9.5 (2.4)	+11.5 (1.2)	+9.4 (0.9)	+11.4 (1.0)	+19.6 (2.2)
Jua EPT-2 Reasoning	+4.6 (0.4)	-6.9 (1.3)	+2.6 (1.3)	+7.4 (0.4)	+7.0 (0.8)	+1.1 (0.7)
Jua EPT-2e	+0.8 (0.7)	-24.3 (3.6)	-5.0 (2.8)	+6.1 (0.8)	+7.4 (1.1)	-5.0 (0.7)
Microsoft Aurora	-1.3 (1.1)	-22.2 (4.4)	-7.0 (4.6)	+4.1 (1.0)	+3.8 (1.6)	-6.9 (1.0)
ECMWF AIFS	-1.4 (0.6)	-27.3 (3.6)	-5.1 (2.3)	+2.3 (1.0)	+4.9 (1.4)	-4.9 (2.0)
ECMWF AIFS ENS (mean)	+0.2 (0.6)	-24.9 (3.2)	-3.1 (2.0)	+4.5 (0.9)	+6.2 (0.9)	-5.1 (1.7)
ECMWF ENS (mean)	-1.7 (0.3)	-12.2 (1.0)	-3.3 (0.9)	+1.4 (0.4)	+0.3 (0.8)	-6.1 (0.5)
NOAA GFS	-23.9 (1.4)	-25.3 (3.5)	-26.3 (2.9)	-26.5 (2.5)	-17.8 (2.1)	-22.8 (2.0)
DWD ICON Global	+5.5 (0.9)	-3.8 (1.2)	+3.0 (1.7)	+4.1 (1.5)	+8.3 (0.9)	+10.2 (2.0)

Table S8: Track A MAE skill (%), 1–48 h hourly – Figure 1, solar. Parentheses: jackknife SE.

Model	All	<P5	P5–25	P25–75	P75–95	>P95
Jua EPT-2.1 Helios	+10.2 (1.7)	+15.4 (5.0)	+16.4 (3.4)	-8.2 (1.1)	+13.7 (2.9)	+24.8 (5.4)
Jua EPT-2.1 Europa	+7.1 (1.0)	+3.1 (4.0)	+11.0 (3.1)	+9.2 (2.3)	+1.9 (4.3)	+4.4 (4.8)
Jua EPT-2 HRRR	+0.6 (1.4)	-3.6 (4.6)	+3.8 (3.7)	+4.5 (1.0)	-4.6 (4.8)	-4.1 (4.9)
Jua EPT-2 Reasoning	+6.3 (0.5)	+0.9 (1.3)	+9.5 (0.8)	+14.0 (0.7)	-0.8 (1.9)	-2.6 (2.6)
Jua EPT-2e	+2.0 (1.4)	-10.6 (3.3)	+5.9 (2.3)	+13.5 (1.6)	-7.5 (4.1)	-7.6 (6.0)
NOAA GFS	-11.2 (1.6)	-3.3 (4.4)	-24.3 (2.1)	-28.3 (2.5)	+8.8 (3.4)	+16.7 (4.2)
DWD ICON Global	+5.6 (0.8)	+6.5 (3.3)	+8.1 (2.6)	+3.1 (1.5)	+4.2 (3.3)	+6.9 (4.2)

Table S9: Track A MAE skill (%), 1–12 h – Figure S5, wind.

Model	All	<P5	P5–25	P25–75	P75–95	>P95
Jua EPT-2.1 Europa	+7.0 (0.5)	-3.7 (1.0)	+3.6 (1.1)	+10.5 (0.7)	+8.1 (0.3)	+8.2 (0.9)
Jua EPT-2 HRRR	+2.0 (0.3)	+0.7 (1.0)	-2.1 (0.8)	+2.4 (0.2)	+3.0 (0.9)	+4.7 (0.9)
Jua EPT-2 Reasoning	+1.5 (0.1)	-1.5 (0.2)	+0.4 (0.3)	+3.2 (0.2)	+3.4 (0.2)	-0.4 (0.2)
Jua EPT-2e	-2.4 (0.2)	-10.8 (0.7)	-6.0 (0.7)	+2.1 (0.5)	+1.7 (0.4)	-6.2 (0.4)
ECMWF ENS (mean)	+0.4 (0.3)	–	–	–	–	–
NOAA GFS	-10.9 (0.8)	+4.8 (0.8)	-5.6 (0.9)	-21.5 (1.8)	-18.1 (1.4)	+0.7 (0.6)
DWD ICON Global	+3.7 (0.3)	-1.3 (0.6)	+4.0 (0.9)	+3.3 (0.7)	+1.5 (0.4)	+8.1 (1.1)

Table S10: Track A MAE skill (%), 1–12 h – Figure S5, temperature.

Model	All	<P5	P5–25	P25–75	P75–95	>P95
Jua EPT-2.1 Europa	+8.3 (0.8)	+2.9 (1.1)	+7.3 (1.0)	+7.8 (1.1)	+7.5 (1.1)	+13.3 (1.4)
Jua EPT-2 HRRR	+9.5 (0.6)	+9.0 (2.3)	+8.7 (1.6)	+6.2 (1.0)	+8.2 (0.8)	+18.8 (2.0)
Jua EPT-2 Reasoning	+2.4 (0.2)	-5.3 (0.6)	+1.0 (0.6)	+4.3 (0.2)	+4.2 (0.5)	+0.0 (0.5)
Jua EPT-2e	-2.5 (0.6)	-24.3 (2.7)	-7.7 (2.2)	+2.3 (0.7)	+3.0 (1.1)	-7.2 (0.5)
ECMWF ENS (mean)	-3.0 (0.2)	–	–	–	–	–
NOAA GFS	-26.9 (1.5)	-29.6 (4.0)	-32.4 (3.5)	-29.0 (2.4)	-20.5 (1.6)	-23.9 (2.5)
DWD ICON Global	+5.3 (0.8)	-2.9 (1.0)	+2.8 (1.2)	+4.9 (1.5)	+6.3 (0.7)	+10.1 (1.4)

Table S11: Track A MAE skill (%), 1–12 h – Figure S5, solar.

Model	All	<P5	P5–25	P25–75	P75–95	>P95
Jua EPT-2.1 Helios	+13.5 (1.9)	+20.6 (5.9)	+20.1 (2.9)	-6.9 (1.5)	+17.7 (2.6)	+28.7 (6.1)
Jua EPT-2.1 Europa	+6.5 (1.4)	+1.2 (3.9)	+10.5 (3.8)	+8.1 (3.0)	+1.9 (4.3)	+5.6 (4.8)
Jua EPT-2 HRRR	-2.1 (1.8)	-6.8 (4.7)	+0.9 (4.3)	+0.3 (1.5)	-6.4 (5.0)	-3.7 (4.8)
Jua EPT-2 Reasoning	+4.4 (0.4)	+0.6 (1.3)	+7.4 (0.8)	+11.0 (0.7)	-2.4 (1.3)	-3.0 (1.7)
Jua EPT-2e	-1.1 (1.7)	-13.5 (3.4)	+4.1 (3.1)	+11.0 (1.7)	-13.8 (4.5)	-10.0 (5.0)
NOAA GFS	-12.7 (1.9)	-5.8 (4.2)	-27.6 (2.7)	-29.8 (2.6)	+9.7 (4.2)	+15.9 (3.6)
DWD ICON Global	+6.4 (1.2)	+6.2 (3.4)	+9.7 (3.0)	+3.4 (1.6)	+4.1 (2.9)	+8.8 (4.8)

Table S12: Track A precipitation MAE skill (%), 1–12 h (Figure S6). Parentheses: jackknife SE.

Model	All	Dry	<P50	P50–75	P75–95	>P95
Jua EPT-2 Reasoning	+1.9 (± 0.5)	-2.9 (± 1.0)	+3.3 (± 1.0)	+8.2 (± 1.0)	+7.8 (± 0.7)	+2.2 (± 0.5)
Jua EPT-2 HRRR	-2.7 (± 1.1)	-10.9 (± 3.8)	+3.9 (± 1.2)	+10.8 (± 1.5)	+6.8 (± 1.1)	-4.5 (± 0.8)
Jua EPT-2e	+4.6 (± 0.9)	+12.4 (± 3.4)	+8.9 (± 1.1)	+11.2 (± 1.1)	+2.2 (± 1.5)	-6.0 (± 1.3)
Jua EPT-2.1 Europa	+2.2 (± 1.5)	+22.0 (± 3.2)	-16.1 (± 4.2)	-16.5 (± 4.6)	-10.8 (± 3.2)	-1.1 (± 0.7)
NOAA GFS	-7.3 (± 1.0)	-3.4 (± 3.6)	-9.1 (± 1.4)	-14.2 (± 1.5)	-16.1 (± 1.9)	-4.8 (± 1.8)
DWD ICON Global	-1.4 (± 2.3)	+11.4 (± 3.3)	-17.4 (± 3.8)	-17.1 (± 4.5)	-11.5 (± 3.5)	-0.3 (± 0.6)

Table S13: Track A MAE skill (%) by lead – Figure S1, wind. Point estimates (no SE in per-lead tables).

Model	6 h		12 h		24 h		48 h	
	All	>P95	All	>P95	All	>P95	All	>P95
EPT-2.1 Europa	+8.3	+8.8	+8.7	+9.3	+8.5	+9.2	+8.1	+9.0
EPT-2 HRRR	+3.3	+5.0	+3.9	+5.6	+4.6	+6.0	+5.1	+6.9
EPT-2 Reasoning	+1.9	-0.7	+2.3	-0.8	+2.9	-0.9	+3.9	-0.4
EPT-2e	-1.2	-8.0	-1.0	-7.5	-0.1	-6.7	+1.1	-5.5
Aurora	-1.6	-4.0	-1.6	-3.6	-0.9	-3.6	+0.4	-2.8
AIFS	-6.6	-10.8	-6.5	-11.0	-5.7	-11.1	-4.0	-9.5
AIFS ENS	-3.5	-8.3	-3.4	-8.9	-2.7	-9.4	-0.7	-8.9
ENS	+0.3	-0.9	+0.7	-0.8	+1.2	-0.9	+2.3	-1.5
GFS	-11.1	+2.0	-11.1	+1.8	-11.8	+0.6	-12.3	-0.5
ICON Global	+4.6	+10.1	+4.6	+10.1	+4.0	+9.6	+2.4	+8.6

Table S14: Track A MAE skill (%) by lead – Figure S2, temperature. Point estimates (no SE in per-lead tables).

Model	6 h		12 h		24 h		48 h	
	All	>P95	All	>P95	All	>P95	All	>P95
EPT-2.1 Europa	+8.4	+12.5	+12.0	+15.8	+11.5	+14.5	+11.6	+14.4
EPT-2 HRRR	+10.1	+19.5	+12.6	+21.3	+12.1	+19.4	+12.2	+18.4
EPT-2 Reasoning	+2.2	-0.5	+3.4	+0.2	+4.5	+0.9	+6.0	+2.2
EPT-2e	-4.2	-10.0	-1.5	-7.5	+1.1	-4.7	+3.2	-2.7
Aurora	-4.6	-7.8	-2.3	-7.1	-1.2	-7.0	+0.2	-6.7
AIFS	-4.5	-7.5	-2.5	-5.2	-1.7	-5.0	+0.4	-3.7
AIFS ENS	-4.3	-10.1	-1.3	-6.6	+0.1	-5.0	+2.2	-3.3
ENS	-3.0	-6.7	-2.6	-6.3	-1.9	-6.0	-0.6	-5.6
GFS	-27.3	-24.6	-24.2	-22.5	-24.2	-23.2	-22.3	-22.6
ICON Global	+5.1	+10.1	+6.9	+11.7	+6.0	+10.2	+4.0	+8.8

Table S15: Track A MAE skill (%) by lead – Figure S3, solar (Ovc. = overcast, P5–P25). Point estimates; SEs in regime tables.

Model	1 h		6 h		12 h		24 h		48 h	
	All	Ovc.	All	Ovc.	All	Ovc.	All	Ovc.	All	Ovc.
EPT-2.1 Helios	+26.1	–	+10.9	+19.0	+9.0	+15.2	+8.1	+15.4	+9.2	+15.5
EPT-2.1 Europa	+6.2	–	+5.6	+10.1	+6.7	+10.0	+6.1	+10.2	+6.3	+10.2
ICON Global	+7.1	–	+5.6	+9.8	+6.2	+8.9	+5.2	+8.1	+4.3	+6.3
EPT-2 Reasoning	+3.5	–	+4.2	+7.0	+5.1	+8.5	+6.0	+9.5	+7.8	+11.1
EPT-2e	+0.8	–	-4.5	+2.4	-2.5	+2.2	+0.2	+2.9	+4.1	+8.1
EPT-2 HRRR	-4.6	–	-4.8	-0.5	-2.7	+1.3	-1.7	+2.6	+1.1	+5.1
GFS	-12.8	–	-13.7	-30.2	-12.4	-29.4	-11.7	-25.8	-11.1	-24.3

Table S16: Track B wind MAE skill (%), 1–48 h, by regime (Figure 5).

Model	All	<P5	P5–25	P25–75	P75–95	>P95
Jua EPT-2.1 Europa	+7.2 (0.4)	+0.1 (0.8)	+3.0 (0.3)	+8.9 (0.6)	+8.2 (0.3)	+10.5 (1.4)
Jua EPT-2 HRRR	+2.4 (0.6)	+4.9 (1.0)	-1.8 (1.7)	+3.5 (0.4)	+1.8 (0.4)	+4.1 (1.5)
DWD ICON-EU	+5.3 (0.5)	-0.7 (1.0)	+1.8 (0.6)	+5.6 (0.5)	+3.9 (0.2)	+11.5 (1.6)

Table S17: Track B MAE skill (%), 1–48 h – Figure 5, temperature.

Model	All	<P5	P5–25	P25–75	P75–95	>P95
Jua EPT-2.1 Europa	+13.2 (0.6)	+5.9 (1.8)	+11.7 (2.0)	+12.3 (1.3)	+13.5 (0.6)	+16.6 (2.3)
Jua EPT-2 HRRR	+12.0 (0.7)	+12.8 (3.3)	+14.9 (1.4)	+9.2 (0.4)	+8.7 (1.3)	+17.2 (3.5)
DWD ICON-EU	+12.4 (0.8)	+7.7 (1.1)	+11.8 (2.3)	+10.6 (1.1)	+11.3 (1.5)	+17.2 (3.2)

Table S18: Track B MAE skill (%), 1–48 h – Figure 5, solar.

Model	All	<P5	P5–25	P25–75	P75–95	>P95
Jua EPT-2.1 Helios	+12.6 (2.5)	+21.6 (4.6)	+18.1 (5.5)	-11.3 (0.5)	+13.5 (5.8)	+32.2 (4.9)
Jua EPT-2.1 Europa	+7.8 (2.0)	+7.4 (3.9)	+12.8 (3.1)	+10.7 (1.7)	+0.1 (4.5)	+3.4 (7.8)
Jua EPT-2 HRRR	+0.8 (2.2)	-0.2 (5.1)	+4.8 (5.7)	+6.5 (0.3)	-6.1 (8.2)	-6.4 (7.7)
DWD ICON-EU	+7.2 (1.4)	+12.4 (3.3)	+12.4 (2.5)	+1.0 (2.2)	+0.9 (3.3)	+10.5 (5.7)

Table S19: Track B MAE skill (%) by lead – Figure S9, wind.

Model	6 h		12 h		24 h		48 h	
	All	>P95	All	>P95	All	>P95	All	>P95
EPT-2.1 Europa	+8.6	+11.1	+9.3	+12.0	+8.9	+11.9	+8.4	+12.3
EPT-2 HRRR	+2.3	+3.8	+3.3	+5.0	+3.9	+5.4	+4.7	+6.7
ICON-EU	+7.5	+14.6	+7.7	+14.7	+7.1	+13.9	+5.4	+13.7

Table S20: Track B MAE skill (%) by lead – Figure S9, temperature.

Model	6 h		12 h		24 h		48 h	
	All	>P95	All	>P95	All	>P95	All	>P95
EPT-2.1 Europa	+10.7	+12.7	+14.6	+17.0	+13.2	+15.1	+13.5	+16.1
EPT-2 HRRR	+12.0	+17.1	+14.2	+19.3	+12.4	+16.8	+12.2	+16.9
ICON-EU	+13.7	+18.4	+14.9	+19.9	+13.2	+17.2	+10.6	+15.5

Table S21: Track B MAE skill (%) by lead – Figure S9, solar (Ovc. = overcast, P5–P25).

Model	6 h		12 h		24 h		48 h	
	All	Ovc.	All	Ovc.	All	Ovc.	All	Ovc.
EPT-2.1 Helios	+12.9	+20.3	+11.4	+17.1	+10.4	+16.4	+11.2	+16.8
EPT-2.1 Europa	+7.2	+13.6	+8.0	+13.5	+6.8	+12.3	+6.9	+11.9
EPT-2 HRRR	-3.5	+3.9	-1.5	+4.7	-1.3	+4.6	+1.1	+6.4
ICON-EU	+7.4	+14.1	+7.8	+13.4	+6.0	+12.0	+4.4	+9.3

Table S22: Track B precipitation MAE skill (%) by lead (Figure S11). Point estimates.

Model	6 h		12 h		24 h		48 h	
	All	>P95	All	>P95	All	>P95	All	>P95
Jua EPT-2.1 Europa	+1.9	+0.5	+9.9	+1.0	+9.0	+1.6	+7.7	+1.9
Jua EPT-2 HRRR	-2.5	-0.6	-2.6	+0.6	-0.9	+1.0	+0.4	+1.7
DWD ICON-EU	-0.1	+0.4	+8.7	-0.6	+7.1	-0.1	+3.2	-1.2

Table S23: Track B MAE skill (%) for 1 h precipitation by wet-hour regime, 1 to 48 h, debiased – Figure S10. Parentheses: jackknife SE.

Model	All	Dry	<P50	P50–75	P75–95	>P95
Jua EPT-2.1 Europa	+7.8 (± 0.9)	+29.3 (± 1.4)	-19.1 (± 6.0)	-20.2 (± 7.2)	-9.0 (± 5.7)	+1.2 (± 0.8)
Jua EPT-2 HRRR	+1.4 (± 3.4)	-3.6 (± 9.7)	+10.3 (± 4.4)	+15.6 (± 6.1)	+10.4 (± 2.5)	-0.7 (± 0.3)
DWD ICON-EU	+3.5 (± 0.5)	+15.3 (± 0.3)	-10.9 (± 2.7)	-11.4 (± 3.8)	-6.9 (± 2.2)	+0.1 (± 0.5)

Table S24: Heavy-precipitation (>P95) detection on Track A, 1 to 48 h, debiased – Figure S7. POD: probability of detection; FAR: false-alarm ratio; CSI: critical success index; frequency bias is forecast over observed event count.

Model	POD	FAR	CSI	Freq. bias
Jua EPT-2.1 Europa	0.28	0.62	0.19	0.74
DWD ICON Global	0.29	0.65	0.19	0.82
Jua EPT-2 Reasoning	0.23	0.56	0.18	0.53
ECMWF IFS	0.25	0.65	0.17	0.71
NOAA GFS	0.25	0.66	0.17	0.74
Jua EPT-2 HRRR	0.20	0.53	0.16	0.42
Jua EPT-2e	0.20	0.55	0.16	0.44
ECMWF ENS (mean)	0.15	0.50	0.13	0.29

A.5 COUNTRY-LEVEL REGIME PROFILES

Figures S12–S15 show how the March–June regime-skill patterns vary across countries for the mixed regional/global comparison set. Countries without sufficient matched observations for a variable are omitted.

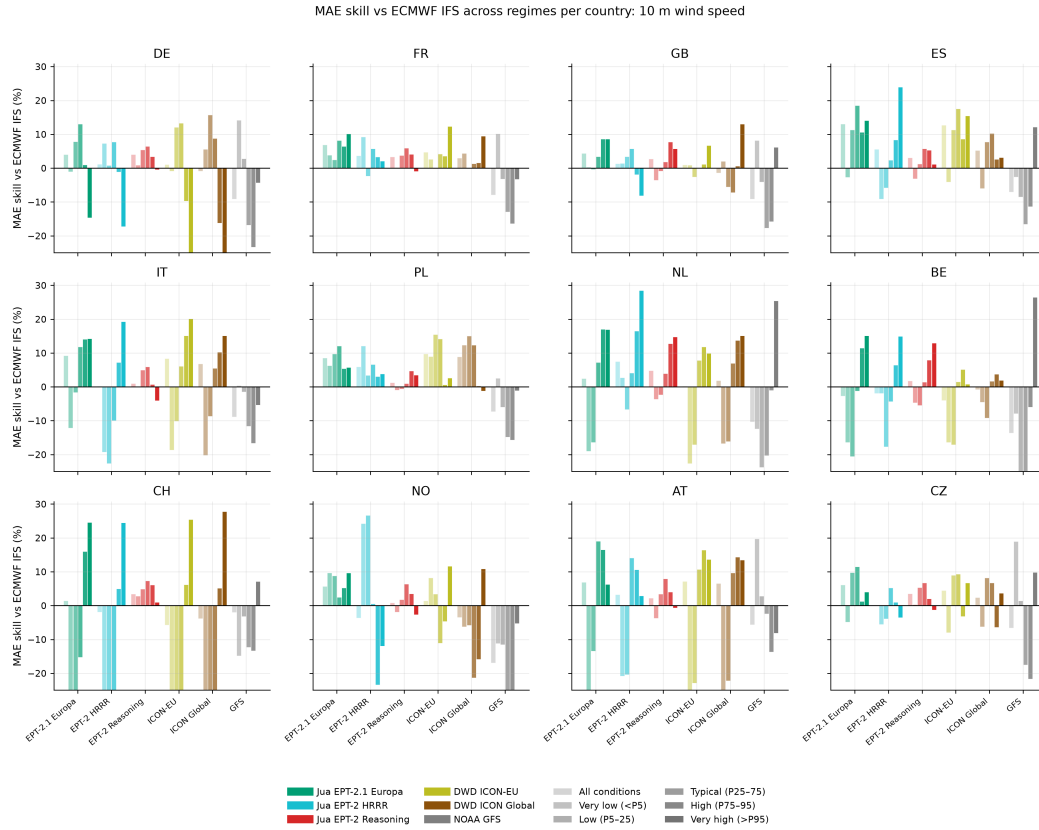


Figure S12: Per-country MAE skill by ERA5 1991–2020 climatological regime, 10 m wind, March–June 2026, pooled 1 to 48 h (hourly). Mixed regional/global model set.

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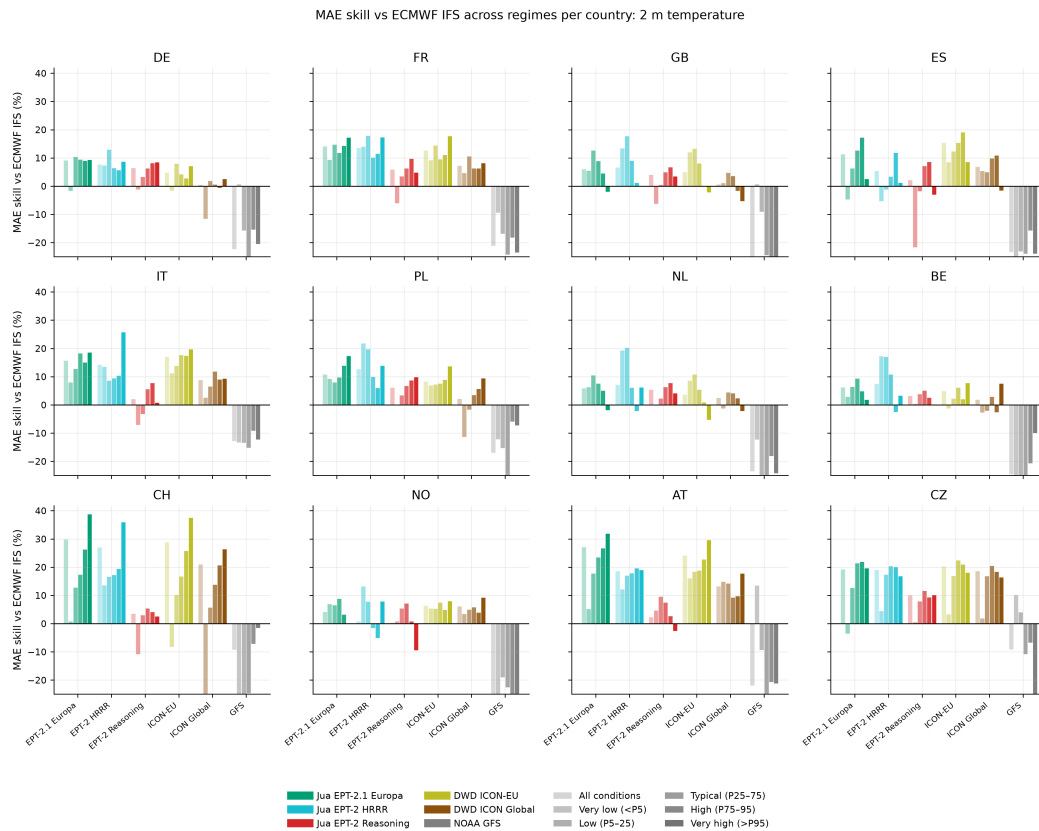


Figure S13: As Figure S12, for 2 m temperature.

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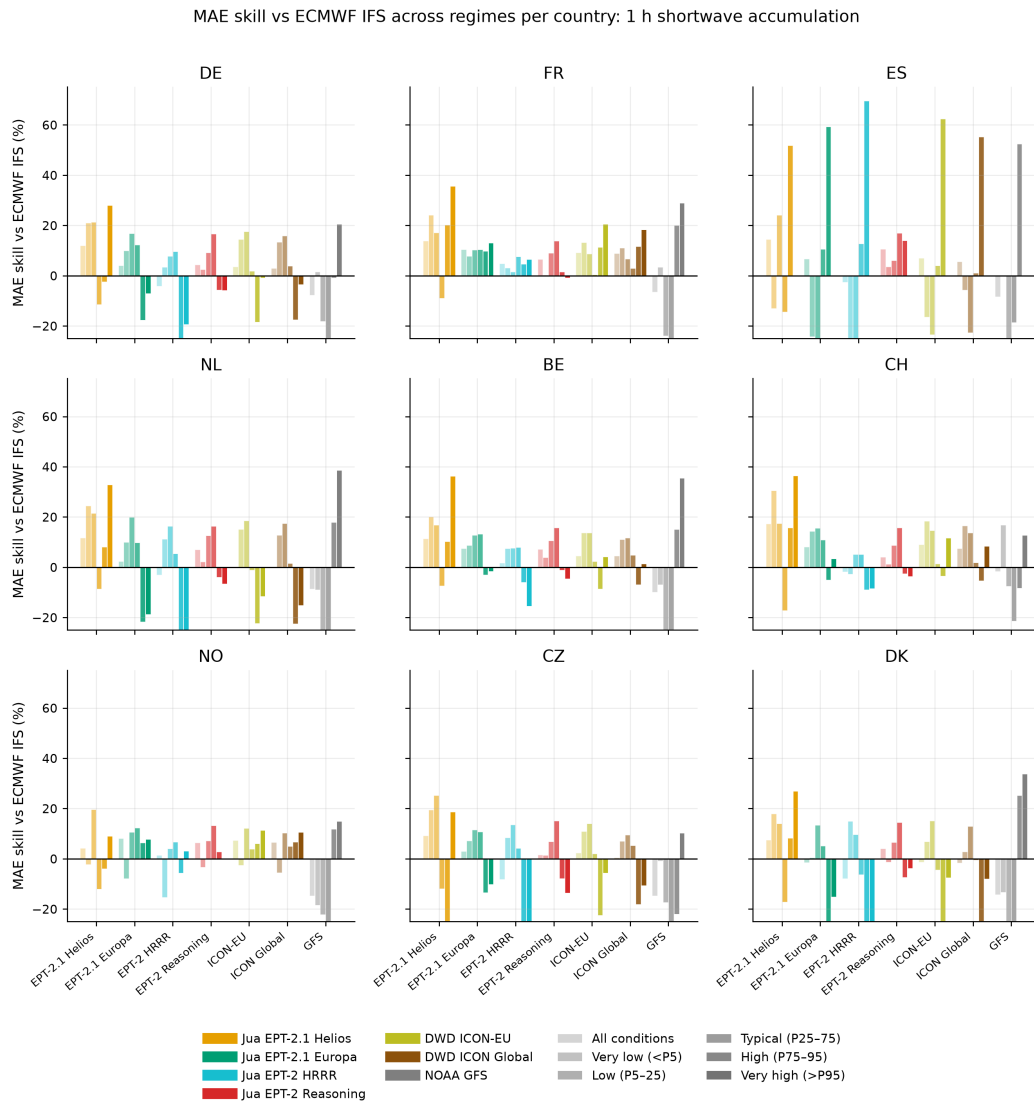


Figure S14: As Figure S12, for 1 h shortwave, with EPT-2.1 Helios added. GB, IT and PL have no solar station observations (Section 2.2).

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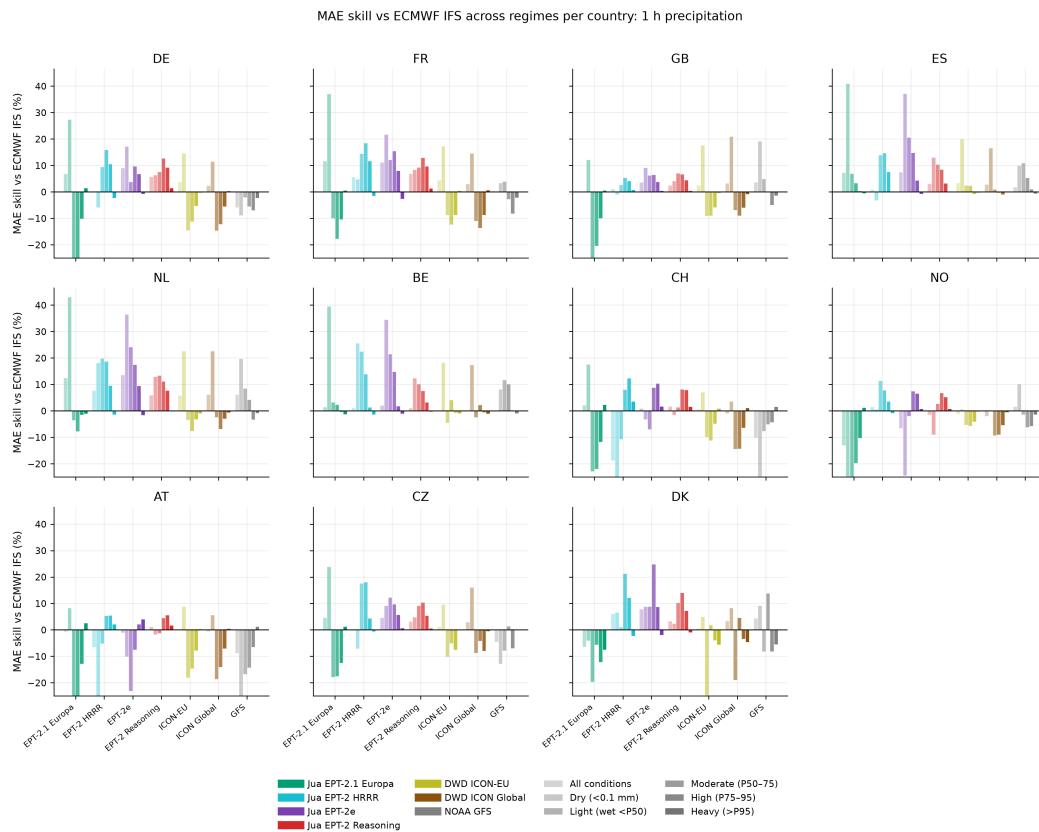


Figure S15: Per-country MAE skill by precipitation regime for the mixed regional/global model set. IT and PL have no gauges in the precipitation panel.

A.6 RECORD-BREAKING TEMPERATURE EVENTS AND MAJOR WINDSTORMS

Table S25 lists the externally documented temperature-record events and European windstorms used in the case-study analysis. Event windows and affected countries were fixed from the source catalogue before scoring. Every benchmark station in the source-reported countries is included at 00, 06, 12, and 18 UTC throughout the event window. The analysis is therefore conditioned on documented events, rather than on whether an individual station observation breaks a record.

For event e and lead $\ell \in \{6, 12, \dots, 48\}$ h, let $\mathcal{C}_{e,\ell}$ denote the station–valid-time pairs (s, t) in the event footprint for which ECMWF IFS and every Track A model included in the comparison provide a forecast initialised at $t - \ell$. Thus, each model is evaluated against the same observations at each lead. For ensemble models, f_m denotes the ensemble mean and $\tilde{f}_m$ the forecast after the causal bias correction of Section 2.4. We compute

$$\begin{aligned}\varepsilon_m(s, t, \ell) &= \tilde{f}_m(s, t | t - \ell) - y(s, t), \\ N &= \sum_e \sum_\ell |\mathcal{C}_{e,\ell}|, \\ \text{MAE}_m^{\text{event}} &= \frac{1}{N} \sum_e \sum_\ell \sum_{(s,t) \in \mathcal{C}_{e,\ell}} |\varepsilon_m(s, t, \ell)|, \\ S_m^{\text{event}} &= 100 \left(1 - \frac{\text{MAE}_m^{\text{event}}}{\text{MAE}_{\text{IFS}}^{\text{event}}} \right).\end{aligned}\tag{5}$$

The same observation may contribute at multiple leads because these are distinct forecasts from different initialisation times. For uncertainty, we delete one complete event at a time and report the jackknife standard error

$$\sqrt{\frac{E-1}{E} \sum_e \left(S_{m,(-e)}^{\text{event}} - \bar{S}_{m,(-\cdot)}^{\text{event}} \right)^2},$$

where $E = 5$ for temperature and $E = 7$ for wind.

Table S25: Source-defined event catalogue. The accompanying reproduction package records the source URLs and selection basis for every event.

Variable	Event	Dates	Affected countries
Temperature	Central European December record warmth	2025-12-07–2025-12-09	NL, CZ
Temperature	Late-February European record warmth	2026-02-25–2026-02-27	FR, DE, ES, CZ
Temperature	Early-April record warm spell	2026-04-05–2026-04-08	FR, GB
Temperature	Western European May heatwave	2026-05-21–2026-05-30	FR, GB
Temperature	Late-June European record heatwave	2026-06-21–2026-06-28	DE, FR, GB, ES, PL, NL, CH, AT, CZ, DK
Wind	North Sea autumn storm	2025-09-15–2025-09-16	NL
Wind	Storm Amy	2025-10-03–2025-10-05	GB, FR, BE, NL, NO, DK, PL
Wind	Storm Benjamin	2025-10-23–2025-10-24	FR, NL, CH, GB
Wind	Storm Goretti	2026-01-08–2026-01-09	GB, FR, DE, NL
Wind	Storm Nils	2026-02-11–2026-02-13	FR, ES
Wind	Storm Marlis	2026-03-25–2026-03-26	CH, DE
Wind	Storm Deborah	2026-03-26–2026-03-27	FR, IT

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Table S26: Event-case MAE skill (%) relative to ECMWF IFS. Parentheses give leave-one-event-out jackknife SE. Temperature uses five events and 231,065 common samples; wind uses seven events and 123,847 common samples.

Model	Temperature events	Windstorms
Jua EPT-2.1 Europa	+15.4 (1.6)	+6.4 (1.8)
Jua EPT-2 HRRR	+12.3 (2.7)	+3.9 (1.3)
Jua EPT-2 Reasoning	+6.5 (0.6)	+3.3 (0.5)
Jua EPT-2e	+4.3 (1.1)	-0.3 (0.8)
Microsoft Aurora	+2.5 (0.9)	+0.7 (0.9)
ECMWF AIFS	-3.4 (3.9)	-6.0 (1.3)
ECMWF AIFS ENS (mean)	+0.1 (2.0)	-5.1 (0.7)
ECMWF ENS (mean)	-0.9 (0.1)	+1.7 (0.7)
NOAA GFS	-25.4 (7.5)	-12.6 (1.8)
DWD ICON Global	+9.8 (2.3)	+2.4 (2.2)